DYLAN CLEMENT BUTELHO

United States (Open to Relocation) | dylanbutelho@gmail.com | +1 (315) 491-1371
dylanbutelho.vercel.app | linkedin.com/in/dylanbutelho | github.com/dylan1011

PROFESSIONAL SUMMARY

Software Engineer and M.S. Computer Science graduate (May 2026) with hands-on experience designing, building, testing, and deploying scalable software systems across full-stack, backend, AI/ML, and cloud domains. Solid command of Python with practical experience integrating LLM APIs (OpenAI, Anthropic) into production applications, architecting prompt engineering pipelines, and building multi-agent coordination systems using LangGraph and LangChain. Experienced building and maintaining RAG systems, contributing to model fine-tuning efforts that improve agent accuracy, and building MERN stack applications (MongoDB, Express.js, React, Node.js) alongside Django and Flask backends. Strong software engineering fundamentals including RESTful APIs, async programming, Docker, Kubernetes, AWS, Terraform, and Git-based workflows. Comfortable partnering with product and engineering to turn user requirements into robust AI-driven features. Leverages AI coding assistants including Claude Code and Codex daily. Authorized to work in the United States. AWS AI Practitioner certified.

EDUCATION

M.S. Computer Science | Syracuse University, Syracuse, NY **Aug 2024 – May 2026**

Coursework: Agentic AI Systems, Design & Analysis of Algorithms, Distributed Systems, Object-Oriented Design, Database Systems, Data Mining

B.E. Information Technology | University of Mumbai, Mumbai, India **Aug 2019 – May 2023**

EXPERIENCE

Software Engineer **Jan 2026 – Present**

iConsult Collaborative | Syracuse, NY

- Architected and iterated on autonomous AI agents using LangGraph and LangChain capable of navigating complex, multi-step environments for document retrieval, query routing, and intelligent response generation across 4 client projects, shipping 6 production features with an 88% first-pass code review acceptance rate.
- Engineered and optimized prompt engineering pipelines and multi-agent coordination systems that power core application features, integrating LLM APIs (OpenAI, Anthropic) with a focus on response quality and reliability, improving AI output quality by 22% across 3 evaluation cycles.
- Spearheaded RAG system development for contextual document retrieval, partnering with product and engineering to turn user requirements into robust AI-driven features deployed with FastAPI and React, maintaining 99.8% service uptime and reducing average API latency by 35% through query optimization and Redis caching.
- Maintained strong software engineering fundamentals through RESTful API development, async programming with Python, Git-based workflows, automated testing with pytest achieving 78% coverage, Docker and Kubernetes deployments on AWS with Terraform IaC, and CI/CD pipelines with Jenkins and GitHub Actions on 2-week agile cycles, reducing deployment cycle time from 90 minutes to under 20 minutes.

Software Engineer **Jan 2024 – Jul 2024**

Vervali Systems | Mumbai, India

- Engineered AI-integrated full-stack applications and automated data pipelines using Python, JavaScript, and TypeScript for a platform with 10,000 active users, processing 50,000+ records daily through Docker-containerized services that reduced deployment cycles by 40%.
- Partnered with product and engineering teams to turn user requirements into delivered features, improving system performance by 35%, maintaining 99.9% data integrity across production systems, and configuring Prometheus and Grafana observability through 6 agile sprint cycles with Jenkins CI/CD.

PROJECTS

OrangeBot: Multi-Agentic AI System *Python · LangGraph · LangChain · FastAPI · React*

- Architected autonomous AI agents with 5 specialized LangGraph agents: 1 planner agent, 5 specialized task agents, and 1 evaluation agent, using ReAct reasoning, MCP tool-calling, and multi-hop retrieval for complex document ingestion, query routing, and response generation.
- Engineered RAG systems for contextual document retrieval using ChromaDB vector store via MCP, architecting prompt engineering pipelines that chain multiple LLM calls (GPT-4o-mini) with structured output parsing and validation for accurate, relevant responses.
- Built the complete backend with FastAPI REST APIs and React frontend, implementing SQLite checkpointer for agent state persistence and Mem0 for long-term memory, enabling agents to maintain context across multi-turn conversations.
- Spearheaded agent accuracy through a structured evaluation pipeline running 500+ test cases across relevance, faithfulness, and hallucination metrics, reducing agent error rates by 28% through systematic prompt optimization.
- Designed the system to run fully locally with no AWS or cloud dependency, using MCP (Model Context Protocol) for tool integration and ensuring all data stays on-premises for privacy and security.

H1B JobPilot: AI Career Navigation Platform ***(In Progress)*** *Python · LangChain · FastAPI · React · Docker · AWS*

- Architected an AI-powered career navigation platform with autonomous agents that ingest USCIS/DOL data and career platform sources for automated job matching, resume tailoring, cover letter generation, and document generation.
- Engineered multi-agent coordination with LangChain for candidate-job matching workflows, closing the loop between AI logic and the user interface with React, processing 20 workflows per day at 99.5% uptime with 80% automated test coverage.
- Built the data ingestion layer scraping career platforms and parsing USCIS H-1B employer data and DOL PERM/LCA disclosure data into a unified PostgreSQL database for intelligent querying and matching.

- Deployed on Cloudflare R2 for object storage with FastAPI backend, React frontend, Docker containerization, and GitHub Actions CI/CD on AWS infrastructure.
- Implemented resume tailoring engine using LangChain that analyzes job descriptions, extracts keywords, and generates tailored resume content matched to each specific role.

LLM Alignment Lab: Multi-Strategy Training Framework *Python · PyTorch · Hugging Face*

- Engineered 4 LLM training methods (SFT, PPO, DPO, GRPO) from scratch in PyTorch on SmolLM2-135M-Instruct, writing actual training loops with gradient computation, loss function optimization, and statistical evaluation rather than using library wrappers.
- Built Supervised Fine-Tuning (SFT) with custom dataset preparation, tokenization, and cross-entropy loss optimization on instruction-response pairs.
- Implemented Proximal Policy Optimization (PPO) with reward model training, advantage estimation, KL divergence penalty, and clipped surrogate objective for reinforcement learning from human feedback.
- Developed Direct Preference Optimization (DPO) bypassing the reward model entirely by optimizing directly on preference pairs using the Bradley-Terry model and log-ratio formulation.
- Created Group Relative Policy Optimization (GRPO) implementing group-based advantage estimation without a separate critic network, reducing computational overhead while maintaining alignment quality.
- Reduced model perplexity by 18% over baseline through systematic benchmarking across BLEU, ROUGE, and perplexity metrics on 500 evaluation samples with iterative optimization. Built for Agentic AI Exhibit at Syracuse University.

ShopGraph: AI-Powered Product Discovery *Python · Flask · React · Pandas*

- Architected a full-stack e-commerce application with React frontend implementing collaborative filtering and Market Basket Analysis using Pearson Correlation Matrix, closing the loop between AI recommendation logic and the user interface for data-driven product discovery.
- Implemented collaborative filtering algorithm that computes user-item similarity scores to generate personalized product recommendations based on browsing and purchase history.
- Built Market Basket Analysis using association rule mining to identify frequently co-purchased products, surfacing cross-sell and upsell opportunities on product pages.
- Designed data modeling layer handling large product datasets with Flask REST API backend, Pandas data processing, and SQL-backed storage for recommendation results and user interaction tracking.

AeroSense: IoT Environmental Monitoring *Python · C++ · Flask · Scikit-learn · Matplotlib · AWS*

- Engineered a real-time IoT monitoring system with C++ sensor data collection (NH3, sulphur, benzene, CO2, O2, humidity, temperature) and Python ML pipeline applying Random Forest classification (94% accuracy) and K-Means clustering to categorize air quality across 6 levels.
- Built the C++ data collection layer interfacing directly with hardware sensors, implementing multi-threaded data acquisition with sub-10ms latency processing 500K records per hour on Linux.
- Analyzed collected sensor data using Python and Flask web application for data visualization and interaction, with Matplotlib dashboards for real-time trend analysis and historical reporting.
- Hosted the application on AWS with EC2 instances for compute, S3 for data storage, and attached observability tools (Prometheus for metrics collection, Grafana for monitoring dashboards) for system health and data pipeline performance tracking.
- Implemented data cleaning and preprocessing pipelines for raw sensor datasheets, handling missing values, outlier detection, and feature engineering before feeding into ML models. Hardware dismantled and AWS decommissioned after project completion.

LeafScan: Deep Learning Diagnostic Engine *Python · TensorFlow · OpenCV · Flask*

- Architected a CNN model trained on 54,000+ images across 38 plant disease classes achieving 96.5% accuracy, using TensorFlow/Keras with convolutional layers, pooling layers, dropout regularization, and softmax classification.
- Built OpenCV image preprocessing pipeline handling user-uploaded plant images with resizing, normalization, color space conversion, and data augmentation to improve model generalization.
- Delivered a Flask web interface for real-time diagnostics allowing users to upload plant images and receive instant disease identification with actionable treatment guidance and confidence scores.
- Implemented model evaluation with confusion matrix analysis, precision/recall per class, and identified the specific failure cases where misclassification occurred for targeted improvement.

E-Commerce Platform: MERN Full-Stack Application *MongoDB · Express.js · React · Node.js*

- Engineered a full-stack MERN e-commerce application with secure user authentication using JWT, product catalog with search and filtering, shopping cart management with real-time updates, and checkout workflow handling end-to-end purchase transactions.
- Designed MongoDB schemas for products, users, orders, and cart management with proper indexing and data relationships, handling concurrent user sessions and inventory management.
- Built Express.js REST APIs for all CRUD operations with middleware for authentication, validation, error handling, and rate limiting.
- Developed responsive React frontend with state management using Context API/Redux, component-driven architecture, and seamless shopping experience across desktop and mobile devices.

Job Portal: MERN Full-Stack Recruitment Platform *MongoDB · Express.js · React · Node.js*

- Architected a dual-interface MERN application with separate dashboards for companies and job seekers, where companies can post jobs, review applications, and select candidates while users browse, filter, and apply to positions with role-based access control.
- Designed MongoDB data modeling for jobs, companies, users, and applications with aggregation pipelines for search and filtering functionality.
- Built Express.js REST APIs with JWT authentication middleware, role-based authorization, and input validation for secure multi-tenant access.

- Developed React frontend with dynamic routing, role-based views, job search with filters (location, salary, category), and application tracking dashboard.

Dating App: MERN Real-Time Matching Platform *MongoDB · Express.js · React · Node.js*

- Engineered a MERN real-time matching application with filter-based user matching algorithms, integrated chat functionality using WebSocket communication, and MongoDB-backed user profiles with secure authentication and session management.
- Built Express.js REST APIs for user management, matching logic, and real-time messaging with Socket.io for bi-directional communication.
- Developed React frontend featuring swipe-based interaction patterns, real-time chat interface, notification system, and responsive design across devices.

Student Classroom Management System *Python · Django · HTML · CSS · SQL*

- Architected a Django web application for educational institutions managing student attendance tracking, academic progress monitoring, student admissions, quiz and examination administration, lecture material distribution, assignment management, and teacher salary administration.
- Designed SQL database schemas for students, teachers, courses, attendance, grades, and administrative records with Django ORM, implementing complex queries for reporting and analytics.
- Built role-based access control for administrators, teachers, and students with different permission levels and dashboard views for each role.

Blogging Platform: Django Content Management System *Python · Django · HTML · CSS · SQL*

- Engineered a Django blogging platform enabling users to create, edit, and publish blog posts with image uploads, providing public blog browsing without authentication for maximum content reach and discoverability.
- Built SQL-backed content management with Django ORM, user authentication system, CRUD operations for blog posts, and responsive HTML/CSS frontend with image handling and rich text editing.
- Implemented SEO-friendly URL routing, pagination, category/tag filtering, and admin dashboard for content moderation.

TECHNICAL SKILLS

AI Agents & LLMs: LangGraph, LangChain, OpenAI API, Anthropic API, Prompt Engineering, Multi-Agent Coordination, RAG, ReAct, MCP Tool-Calling, Claude Code, Codex

ML & Training: PyTorch, TensorFlow, Hugging Face, SFT, PPO, DPO, GRPO, Scikit-learn, OpenCV, CNN, Random Forest, K-Means, Model Evaluation, BLEU, ROUGE, Pandas, NumPy, Matplotlib

Languages & APIs: Python, JavaScript, TypeScript, Java, Go, C, C++, C#/.NET, SQL, Bash, RESTful APIs, GraphQL, Async Programming, FastAPI, Flask, Django, Celery, WebSocket

Full Stack: React, React Native, Node.js, Express.js, Next.js, MongoDB, PostgreSQL, MySQL, SQL Server, Redis, Supabase, HTML/CSS, D3.js

Infrastructure: Docker, Kubernetes, AWS (EC2, S3, RDS, Lambda, CloudWatch, Bedrock), GCP, Cloudflare, Terraform, Ansible, Git, GitHub, GitHub Actions, Jenkins, CI/CD, Prometheus, Grafana, Datadog, OpenTelemetry, n8n, Linux/Unix, Agile/Scrum, Jira

Tools & Design: AutoCAD, Figma, Canva, Tableau, Power BI, Excel, Shopify, WordPress, Wix

CERTIFICATIONS

AWS AI Practitioner (AIF-C01) | AWS Developer - Associate *(In Progress)*

Cisco Python Essentials I & II | **Cisco Data Science Essentials**